

Aniline Blue Staining

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Purpose. Pollen-tube visualization in pistils, ovules, and seeds.

Pistil workflow

1. Collect pollinated pistils, for example 24 h after pollination.
2. Fix briefly in ethanol:acetic acid (3:1) for 2 h at room temperature.
3. Wash three times with distilled water and soften in 8 M NaOH overnight.
4. Wash with D.W. and stain in 0.1% aniline blue in 0.1 M $\text{K}_4\text{Fe}(\text{CN})_6 \cdot 3\text{H}_2\text{O}$ buffer, pH 11, for 3-5 h in the dark.

Ovule workflow

1. Fix pollinated pistils in ethanol:acetic acid (9:1) for several hours.
2. Treat with 1 M NaOH for approximately 16 h.
3. Stain in 0.1% (w/v) aniline blue in 0.1 M K_3PO_4 for more than 3 h.